Module 10 (c)
Item and Test Analyses Based on Item Response Theory

Test Theory	Psychometric Property	Number of Dimensions or Type of Validity	Item Type or Validity Subtype	Possible Techniques
Item Response Theory (IRT)	Descriptive item and test statistics	Unidimensional	Dichotomous	1PL, 2PL, 3PL
			Polytomous	Graded Response Model
				Partial Credit Model
				Rating Scale Model
			Mixed	Mixed-Format IRT Models
		Multidimensional		Multidimensional IRT Models
	Reliability	Unidimensional		Item Information Function
				Test Information Function
		Multidimensional		Multidimensional Item Information Function
				Multidimensional Test Information Function
	Validity	Content		Content Validity Index
				Concept Mapping
				Q-Sort Method
				Item Parameters
		Criterion-related		Latent Trait Correlation
				Item Information Function
				Test Information Function
		Discriminant		Likelihood Ratio Test Between Nested Models
				Item Characteristic Curve
				Item Discrimination Parameter
		Construct	Structural	Item Fit Analyses
				Multidimensional IRT Model
			Differential item functioning	Lord's Chi-Square
				Raju's Area Measures
				SIBTEST
				TSW Likelihood Ratio
			Test equating	Holland and Rubin
				Haebara
				Lord
				Stocking and Lord
			Measurement invariance	Multigroup IRT Modeling